

SURNAME

NAME

MAT. NUMBER

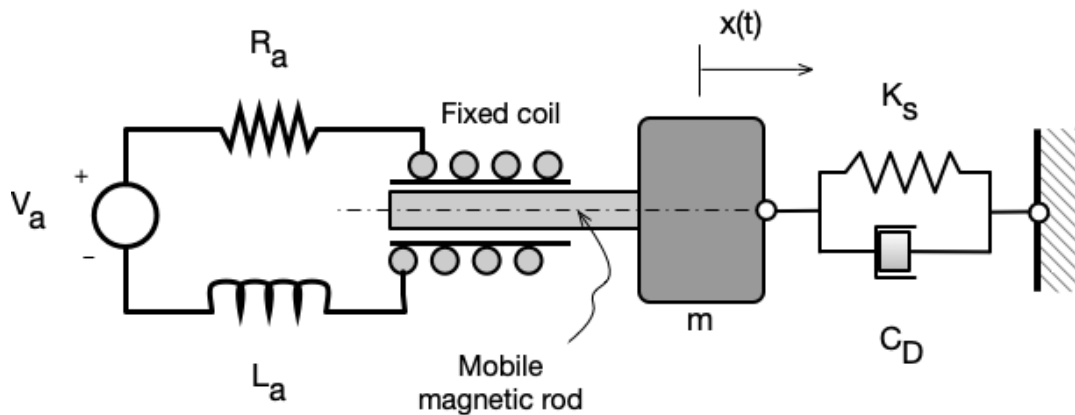
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1 Modelling: Exercise 1

Figure 1 shows the schematic representation of a DC linear-motion motor with its actuation circuit and mechanical load subsystems. The interaction between the current passing through a fixed cylindrical coil and a magnetic rod that can slide inside the coil generates an actuation force $f_a(t)$, which is used as input to the a mass-damper-spring mechanical load system. A back-electromotive force (voltage) $v_b(t)$ is also generated in the actuation electrical circuit, which opposes the actuation voltage $v_a(t)$. The following equations are the relationships that realize coupling between the electromagnetic and mechanical fields:

$$f_a(t) = K_m i(t), \quad v_b(t) = K_{\text{bemf}} \frac{dx(t)}{dt}$$

where K_m is the force constant and K_{bemf} is the back-emf constant. The motor coil has an inductance L_a and a resistance R_a .



Respond to the following questions:

- 1 Draw the corresponding linear graph and the normal tree.
- 2 write the incident matrix and derive the **circuit equations** and the **node equations**.
- 3 Write the list of **constitutive equations** for each element in the hydraulic-mechanical system.
- 4 Assemble the equations and specify the list of input variables, the list of system states and the list of equations of dynamics.

Report in the corresponding boxes the final result of what is required. All the calculation done to arrive to the final solution can be derived manually on give paper or using Matlab Symbolic Toolbox or Maple. The paper with manual detailed solution must be handed to the examiner and the Maple or Matlab file must be uploaded to the Moodle platform.

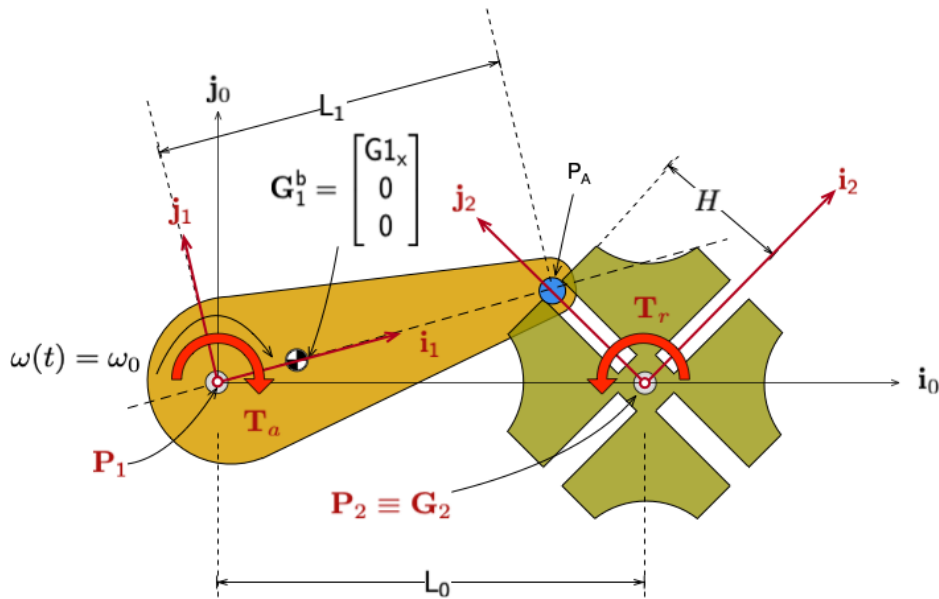
Draw the corresponding Linear Graph (LG) and the normal tree.:

Report the incidence matrix.

Report the final set of equations.

2 Modelling: Exercise 2

The figure 2 shows a *Geneva mechanism* which is a gear mechanism that translates a continuous rotation movement into intermittent rotary motion. The rotating drive body contains a pin P_A that reaches into a slot of the driven wheel (i.e. the cross-shaped body) advancing it by one step at a time. The driven wheel has four slots of length H along the unit vectors of its reference frame (i.e. $\mathbf{i}_2, \mathbf{j}_2$). The drive body rotates with a clockwise constant angular velocity ω_0 around the fixed point $\mathbf{P}_1$. The driven wheel rotates counter-clockwise around the fixed point $\mathbf{P}_2$ as shown in figure. **We are interested in modelling the system only when the pin is in contact with the driven wheel.** The drive body is subject to a torque $T_a(t)$ and the driven wheel is subject to a load torque $T_r(t)$. The drive body has a mass m_1 and a moment of inertia J_1 around the point $\mathbf{G}_1$ (its centre of mass). The driven wheel has a mass m_2 and a moment of inertia J_2 around the point $\mathbf{P}_2$. The distance between the two rotation points $\mathbf{P}_1$ and $\mathbf{P}_2$ is L_0 . The gravity force acts along the negative direction of the unit vector $\mathbf{j}_0$. Respond to the following



questions:

- 1 Draw the LG and the normal tree of the system
- 2 derive the kinematic constraint equations using the recursive formulation (i.e. loop equations)
- 3 derive the expression of conditions of pin P_A engaged with wheel.
- 4 derive the expression of the point $\mathbf{G}_1$ and its velocity and acceleration.
- 4 derive the node equations and the equations of motion using the Neutown Euler approach
- 5 *Optional* include the friction in the contact bewteen the slot and the pin P_A .

Suggestion: consider the rods always in contact with the L-shaped lever arm and also use the refernce frames defined in the picture.

Report in the corresponding boxes the final result of what is required. All the calculation done to arrive to the final solution can be derived manually on give paper or using Matlab Symbolic Toolbox or Maple. The paper with manual detailed solution must be handed to the examiner and the Maple or Matlab file must be uploaded to the Moodle platform.

Draw the corresponding Linear Graph (LG) and the normal tree.:

Report the loop equations.

Report the node equations.

Report the final set of equations of motion.